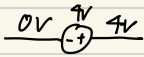




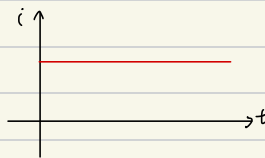
◦ Voltage, V (Joules/Coulomb or Volts)

- Defn 1: $\frac{\text{Change in Potential energy}}{\text{Charge}} = \frac{dw}{dq}$

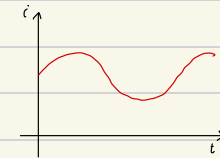
- Defn 2: Difference in electric potentials, $\Delta V = V_{AB} = V_A - V_B$

- Voltage rise: When we move from -ve to +ve terminal $\rightarrow$ 

◦ DC current:



◦ AC current:



◦ Charge, q (Coulombs)

- Defn: electrical property of particles

- Charge of electron (e) = 1.602×10^{-19} C $\Leftrightarrow$ 1C of charge = $\frac{1}{1.602 \times 10^{-19}}$ electrons.

- Charge transferred between t_0, t_1 : $Q = \int_{t_0}^{t_1} i dt$

◦ Current, i (Coulomb/seconds or Amps)

- Defn: rate of change of charge, $i = \frac{dq}{dt}$

◦ $E \rightarrow$ Electric field.

$B \rightarrow$ Magnetic field.

◦ Power, P (Joules/second or Watts)

◦ Tellegen's Theorem / Conservation of energy

- $P = vi$ and energy = $\int_{t_0}^{t_1} p dt$ [J]

$\hookrightarrow$ Power supplied by cct = Power absorbed by cct or $\sum p = 0$

- +ve power = absorbing energy

- -ve power = delivering energy

$\hookrightarrow$ Ohm's law: $P = i^2 R$ or $P = \frac{v^2}{R}$

(independent)

◦ Resistors



$\hookrightarrow$ Element that restricts charge flow

$\hookrightarrow$ No polarity, always absorb power

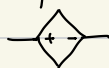
◦ Ideal / Dependent source

$\hookrightarrow$ Ideal:



, provides set voltage/current regardless of connection

$\hookrightarrow$ Dependent source:



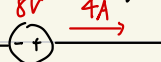
, provides voltage/current depending on another voltage or current



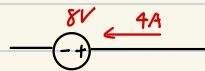
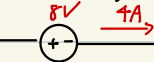
◦ Passive sign convention:

$\hookrightarrow$ If a positive current flows out of the positive side of a voltage, the element delivers power
Else, it absorbs power

$\hookrightarrow$ Delivering



$\hookrightarrow$ Absorbing

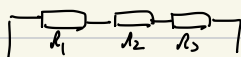


◦ Ohm's Law

$$V = IR, \quad I = \frac{V}{R}, \quad R = \frac{V}{I}$$

◦ Equivalent Resistance

- Series:



$$R_{eq} = R_1 + R_2 + R_3$$

- Parallel:



$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)^{-1}$$

◦ Resistance (Ohms, Ω):

$$R = \frac{\rho L}{A}$$

Resistivity: ρ
Length: L

(Surface) Area: A

◦ $1kWh = 3.6 MJ$

Conductance, G (Siemens, $\mathcal{S}$)

↳ Measurement of how easily the object allows current to flow

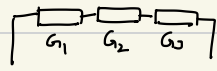
$$G = \frac{1}{R} = \frac{A}{\rho L} = \frac{\sigma A}{L}, \quad \text{Conductivity: } \sigma = \frac{1}{\rho}$$

↳ Thus,

$$G = \frac{I}{V}, \quad P = \frac{I^2}{G} = V^2 G$$

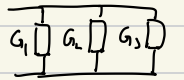
Equivalent Conductance

↳ Series:



$$G_{eq} = \left(\frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3} \right)^{-1}$$

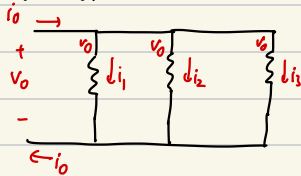
↳ Parallel:



$$G_{eq} = G_1 + G_2 + G_3$$

Parallel / Series Circuit

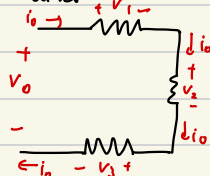
Parallel:



$$i_0 = i_1 + i_2 + i_3$$

$v_0 = \text{equal at all branches}$

Series:



$$i_0, \text{ equal throughout}$$

$$v_0 = v_1 + v_2 + v_3$$

Nodes, Branches, Loops

↳ Branches, a single element

↳ Nodes, a point of connection between 2 or more branches

↳ Loops, closed path in a circuit,

Fundamental theorem of network topology:

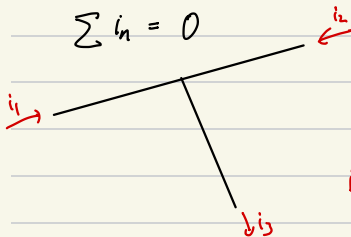
$$b = l + n - 1, \quad l \text{ is the \# of independent loops}$$

Kirchhoff's Laws

↳ KCL:

Current entering node = Current leaving node or

$$\sum i_n = 0$$



$$i_1 + i_2 = i_3$$

or

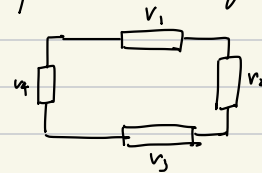
$$i_1 + i_2 + (-i_3) = 0$$

↳ KVL:

Sum of all voltages in a loop = 0 /

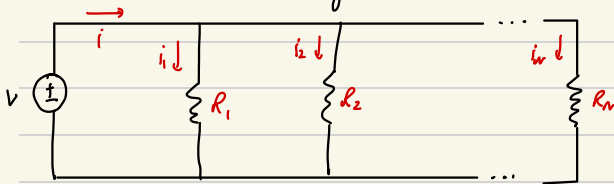
Sum of voltage drop = Sum of voltage rise. or

$$\sum V_n = 0$$

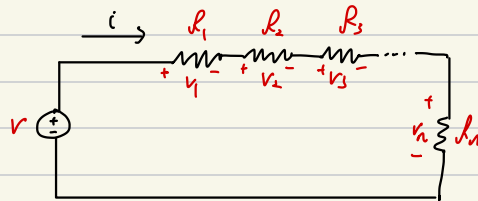


$$v_1 + v_2 + v_3 + v_4 = 0$$

Current and Voltage division



$$i_n = \frac{G_n}{G_1 + G_2 + \dots + G_n} (i)$$

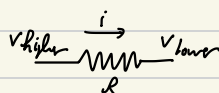


$$v_n = \frac{R_n}{R_1 + R_2 + \dots + R_n} (V)$$

Nodal Analysis.

↳ $N-1$ equations

$$i = (v_{\text{higher}} - v_{\text{lower}}) / R$$



↳ Steps to solve super nodes

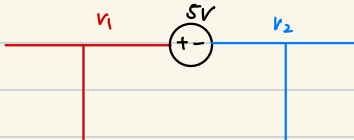
1. Find constraint eq. $\rightarrow v_1 = v_2 + 5V$

2. Apply KVL / KCL (w/o involving the source)

↳ Super-nodes, occurs when a (dependent/independent)

voltage source is connected between two

non-reference node (voltage not ground $\rightarrow 0V$ or unknown value)



Mesh analysis

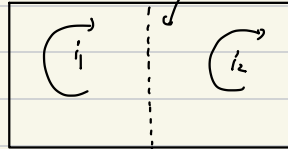
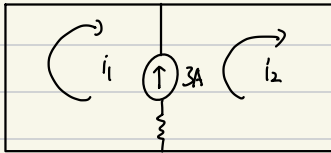
↳ N equations (1st window panes)

↳ Supermesh, occurs when 2 meshes have a (dependent/independent) current source in common.

↳ Solving supermesh

1. Set up constraint eq. $\rightarrow (-i_1) + i_2 = 3A$

2. Consider the whole mesh



Nodal and mesh analysis by inspection.

↳ Only applicable when all sources are independent

Nodal:

$$\begin{bmatrix} G_{11} & G_{12} & \dots & G_{1N} \\ G_{21} & G_{22} & \dots & G_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ G_{N1} & G_{N2} & \dots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix}$$

Mesh:

$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1N} \\ R_{21} & R_{22} & \dots & R_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1} & R_{N2} & \dots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

G_{kk} = Sum of conductance connected to node k

$G_{kj} = G_{jk}$ = -ve of sum of conductance directly connecting nodes k, j

v_k = Unknown voltage @ node k

i_k = Sum of independent sources connected to node k , entering node = +ve.

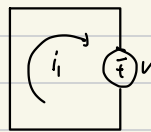
R_{kk} = Sum of resistance in mesh k

$R_{kj} = R_{jk}$ = -ve of sum of resistance shared between nodes k, j

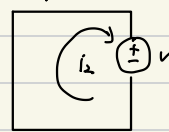
i_k = Unknown current @ mesh k

v_k = Sum of all voltage source in a mesh, voltage rise taken as +ve (rise versa)

Rise:



Drop:

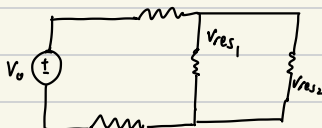


Circuit linearity

↳ $k i R = k v$

i.e. If the circuit is linear,
scaling v by k also
scales i by k

(Applies to any part of the circuit)



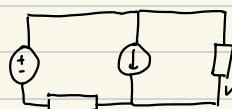
If v_0 is divided by 5,

v_{res1} and v_{res2} must also
be divided by 5.

Superposition

↳ Switch off all except one independent source
and find the "contribution" of the value

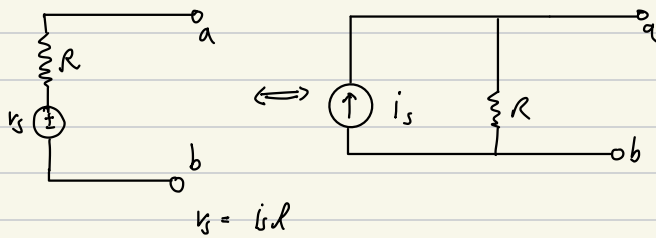
↳ Dependent sources are kept on



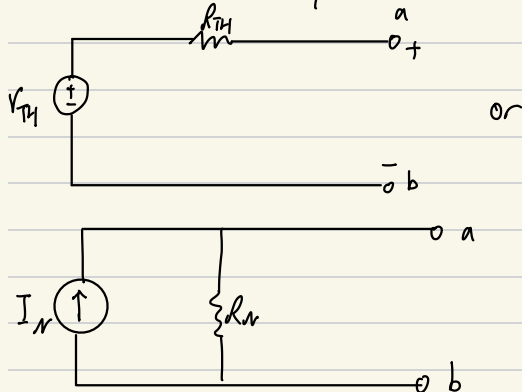
$$v = v_1 + v_2$$



Source transformation



Theremin / Norton Equivalent



- To find R_{TH} :

When there are only independent sources:
 $\hookrightarrow$ Turn off all sources, find R_{eq}

- To find V_{TH} :

$\hookrightarrow$ Solve the circuit dumbass

When there are only dependent sources:

$\hookrightarrow$ Apply test voltage (or current)
 find $R = \frac{V}{I}$

- To find I_N :

$\hookrightarrow$ Solve the circuit dumbass

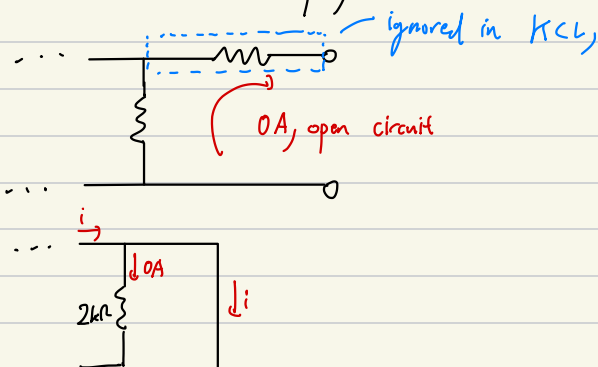
When there are both independent and dependent sources:

$\hookrightarrow$ Find both open circuit voltage and short circuit current
 find $R_{TH} = V_{oc} / I_{sc}$

- Tips on finding V_{TH} :

- Redraw the circuit.

- Be sure what is open, what is shorted.



All current goes through short cct.

- Key points:

$\hookrightarrow R_{TH} = R_N$

$\hookrightarrow V_{TH} = R_{TH} \text{ (or } R_N) \cdot I_N$

$\hookrightarrow V_{TH} = 0, I_N = 0$ for circuits with only dependent sources.

But $R_{TH} \neq 0$

Maximum Power Transfer.

$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} \quad \text{or} \quad \frac{I_N^2 R_{TH}}{4} \quad \text{where}$$

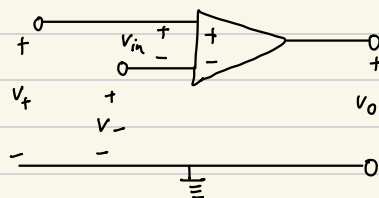
load resistor = theremin resistance



Op-Amps

↳ Acts like a voltage controlled voltage source

↳ Input to Op-Amp = $V_+ - V_-$



$$A_o = \frac{V_o}{V_{in}}$$

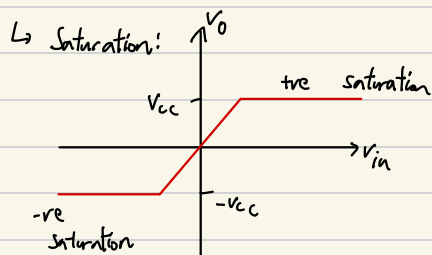
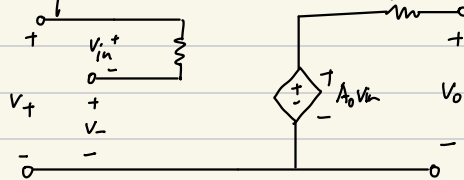
Properties of Op-Amps:

↳ Large "input" resistance

↳ Small "output" resistance

↳ Very large voltage gain

Equivalent circuit:



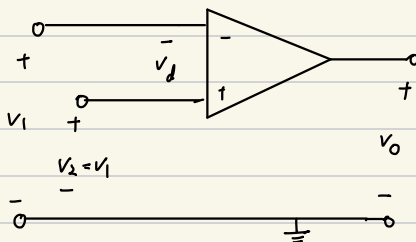
Ideal Op-Amp:

↳ Assumes

1. $A \approx \infty$

2. $R_i \approx \infty$

$R_o \approx 0$

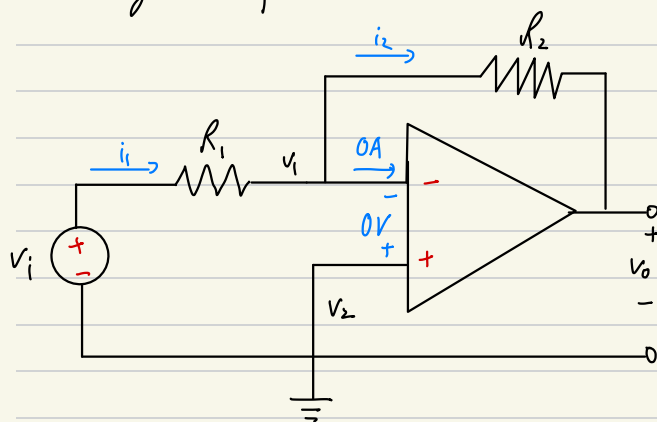


↳ Thus,

1.) Input current, $i_1, i_2 = 0$

2.) Input V is equal, $V_1 = V_2$
 $V_d = 0$

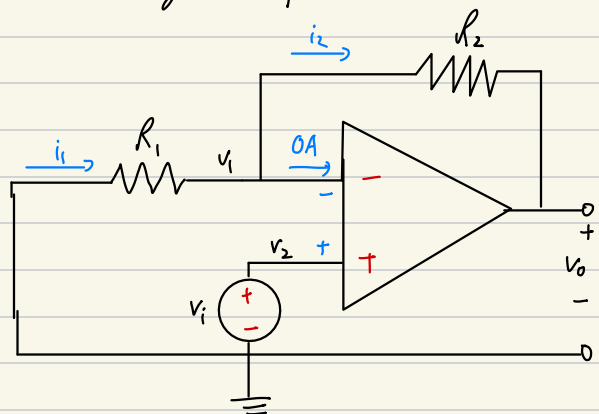
Inverting Amplifier:



$$V_o = -\frac{R_2}{R_1} V_i, \text{ Voltage gain} = -\frac{R_2}{R_1}$$

↳ Reverses polarity, and amplifies it

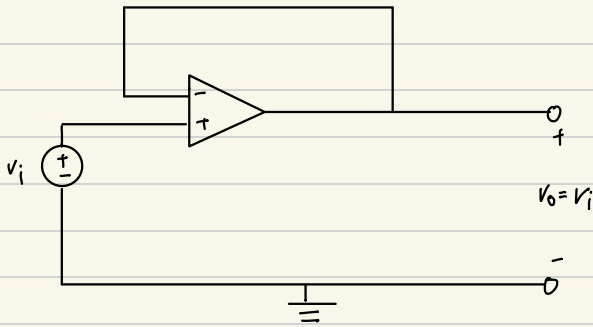
Non-inverting Amplifier



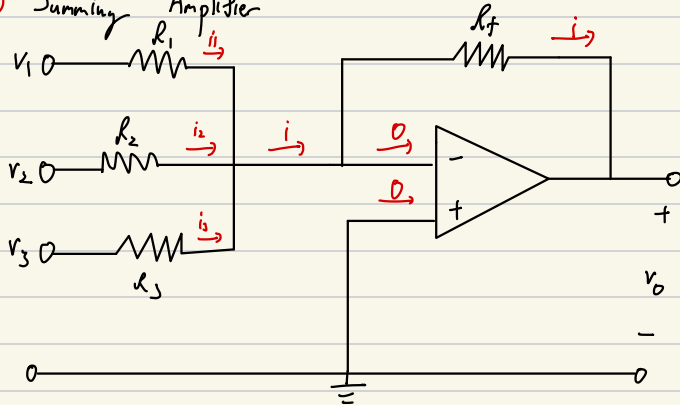
$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i, \text{ Voltage gain} = 1 + \frac{R_2}{R_1}$$

↳ Provide the voltage gain, never < 0 , $\frac{R_2}{R_1} \geq 0$

- Unity-gain buffer amplifier / voltage follower
 - ↳ Useful for isolating one circuit from another



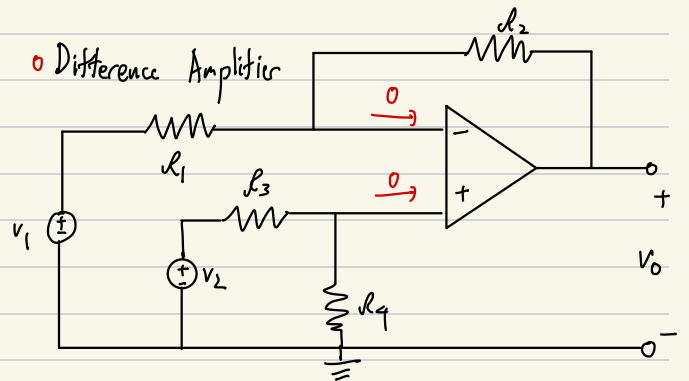
◦ Summing Amplifier



$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right),$$

the output is a "weighted sum" of inputs

◦ Difference Amplifier

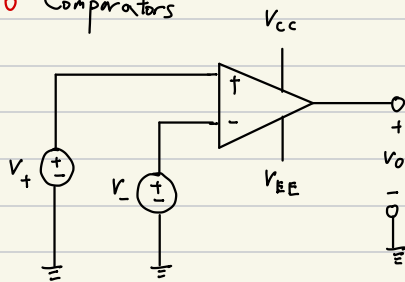


$$v_o = \frac{R_2 (1 + R_1/R_2)}{R_1 (1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1 \rightarrow \text{If } R_1=R_2, R_3=R_4$$

$$= \frac{R_2}{R_1} (v_2 - v_1),$$

the output is the difference of voltage sources

◦ Comparators



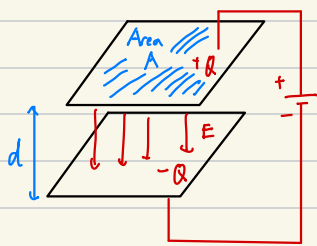
- No feedback connection b/w output, input
"open loop configuration".

- Utilized to "compare" inverting, non-inverting input voltages

$$v_o = \begin{cases} V_{cc} & v_+ > v_- \\ V_{EE} & v_- > v_+ \end{cases}$$

Capacitor, Capacitance:

↳ Parallel-Plate Capacitor



↳ Separation of charge creates
 E , electric field $= \frac{V_0}{d}$ — Applied voltage

↳ E strength:

$$E = \frac{Q}{A\epsilon_r\epsilon_0} \quad \text{— permittivity of free space}$$

relative permittivity $\sim 8.85 \times 10^{-12} [F/m]$

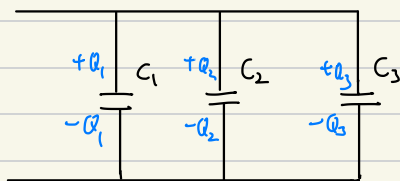
↳ Capacitance, $C = \frac{Q}{V_0} = \frac{A\epsilon_r\epsilon_0}{d}$
 in F, Farads $\left[\frac{\text{Coulombs}}{\text{volts}} \right]$

↳ Energy stored, $W = \frac{1}{2} C V_0^2 = \frac{1}{2} \frac{Q}{V_0} = \frac{1}{2} \frac{Q^2}{C}$

↳ $V_C(t)$ cannot change abruptly
 $i_C = C \frac{dV_C}{dt} \neq \infty$

Equivalent Capacitance:

↳ Parallel:

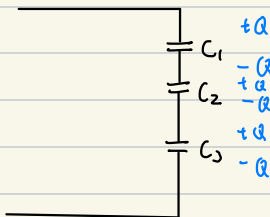


↳ Same voltage
 Different charges

$$C_{eq} = C_1 + C_2 + C_3, \quad Q_1 = C_1 V, \quad Q_2 = C_2 V, \quad Q_3 = C_3 V$$

↳ A capacitor is an **open circuit** when it is **dc**
 (after a long time, $t \rightarrow \infty$)

↳ Series:



↳ Same charges
 Different voltages

$$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)^{-1}, \quad V_1 = \frac{Q}{C_1}, \quad V_2 = \frac{Q}{C_2}, \quad V_3 = \frac{Q}{C_3}$$

Current, Power and energy in capacitor

↳ Current:

$$i_C(t) = C \left(\frac{dV_C}{dt} \right)$$

↳ Voltage

$$V_C(t) = \left[\frac{1}{C} \int_0^t i_C(\tau) d\tau \right] + V_C(0)$$

↳ Power:

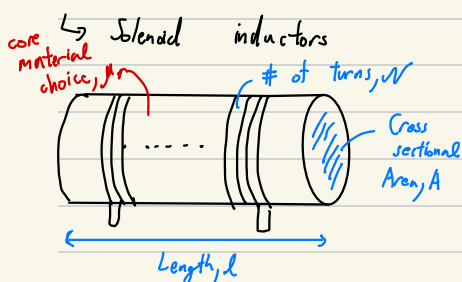
$$p_C(t) = V_C(t) i_C(t)$$

— discharging when -ve
 — absorbing when +ve

↳ Energy:

$$W_C(t) = \frac{1}{2} C V_C^2(t) = \int_{-\infty}^t p_C(\tau) d\tau = \frac{q^2}{2C}$$

Inductors, inductance



↳ Magnetic flux density, B [Tesla, T]

$$B_{inside} = \left(\frac{\mu_r \mu_0 N}{l} \right) i, \quad B_{outside} = 0$$

relative permeability

↳ Inductance, $L = N \frac{\Phi_m}{i} = \frac{\mu_r \mu_0 N^2 A}{l}$

in Henries, H $\left[\frac{\text{Webers, Wb}}{A} = \frac{Vs}{A} \right]$
 ($\times N$) as each coil experiences (DC)

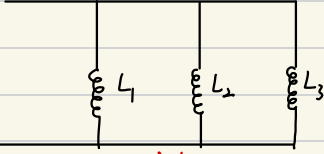
↳ Magnetic flux, $\Phi_m = (B)(A)$
 $= \frac{\mu_r \mu_0 N A}{l} i$

↳ Current in an inductor cannot change instantaneously,
 $v_L = L \frac{di_L}{dt} \neq \infty$

— Magnetic field outside the solenoid is ideally 0.

Equivalent inductance

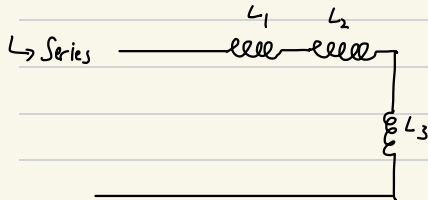
↳ Parallel:



↳ Same voltage, different currents

↳ An inductor is a short circuit when it is dc (after a long time, $t \rightarrow \infty$)

$$L_{eq} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right)^{-1}$$



↳ Same currents, different voltages

$$L_{eq} = L_1 + L_2 + L_3$$

Voltage, Power and energy in inductor

↳ Voltage:

$$v_L(t) = L \frac{di_L}{dt}$$

↳ Current:

$$i_L(t) = \left[\frac{1}{L} \int_0^t v_L(\tau) d\tau \right] + i_L(0)$$

↳ Power:

$$p_L(t) = v_L(t) i_L(t)$$

↳ Energy:

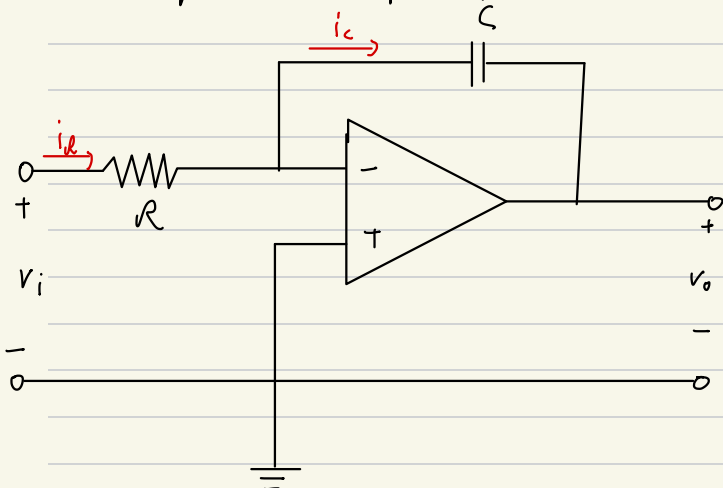
$$W_L(t) = \frac{1}{2} L i_L^2(t) = \int_{-\infty}^t p_L(\tau) d\tau$$

- discharging when -ve

- absorbing when +ve

Integrator

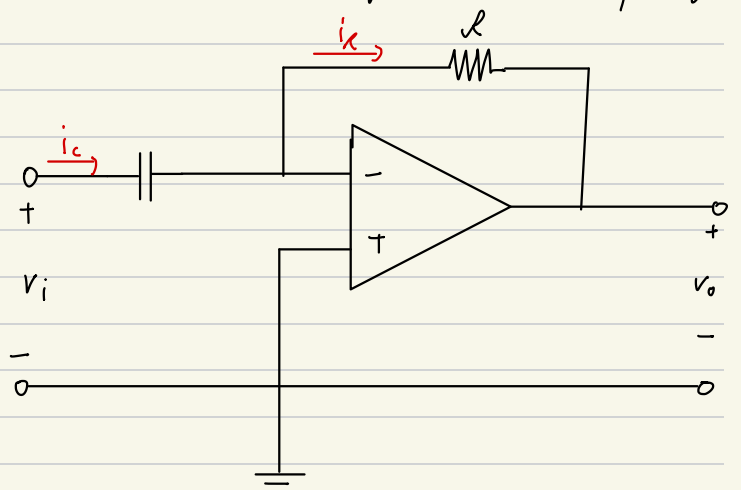
↳ An op amp circuit whose output is proportional to the integral of the input signal



$$V_o = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$

Differentiator

↳ An op amp circuit whose output is proportional to the rate of change (derivative) of input signal



$$V_o = -RC \frac{dv_i}{dt}$$

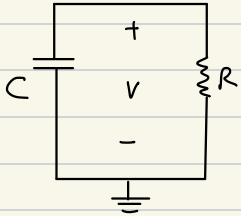
Recap of first-order ODE

For $v' + P(t)v = f(t)$

Integrating factor = $\bar{p}(t) = e^{\int P(t) dt}$

Solution, $v(t) = \frac{1}{\bar{p}(t)} \int \bar{p}(t) f(t) dt + C$

Source-Free RC Circuit



$\hookrightarrow v(t) = V_0 e^{-t/\tau}$

$\hookrightarrow \tau$, time constant = RC

$\uparrow \tau \rightarrow$ time it takes for response to decay to $1/e$ (or 36.8%) of initial value
 $5\tau \rightarrow$ steady state

$\hookrightarrow$ "Source free"

$\hookrightarrow$ "sudden disconnection of source"

$\hookrightarrow V_c(0^-) = V_c(0^+)$

voltage cannot change instantaneously

$\hookrightarrow$ Before switch closed:

"open circuit"

Capacitor
 $\hookrightarrow i_c(t) = C \frac{dv_c(t)}{dt}$, current

$\hookrightarrow p_c(t) = v_c(t) i_c(t)$, power

$\hookrightarrow w_c(t) = \frac{1}{2} C v_c^2(t)$, energy

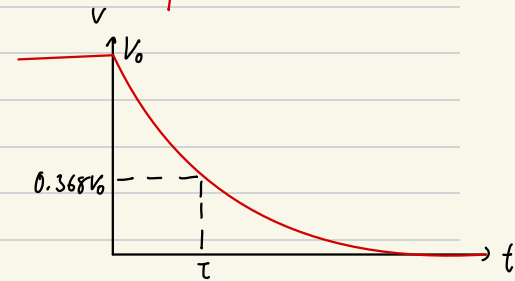
Resistor:

$\hookrightarrow i_R(t) = \frac{v_R(t)}{R} = \frac{V_0}{R} e^{-t/\tau}$, current

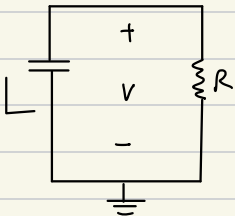
$\hookrightarrow p_R(t) = v_R(t) i_R(t) = \frac{V_0^2}{R} e^{-2t/\tau}$, power

$\hookrightarrow w_R(t) = \int_0^t p_R(t) dt = \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau})$

$= \frac{1}{2} C V_0^2$ for large t , energy



Source-Free RL Circuit



$i(t) = I_0 e^{-t/\tau}$

$\hookrightarrow \tau = \frac{L}{R}$

$\hookrightarrow i_L(0^-) = i_L(0^+)$

"current cannot switch instantaneously"

$\hookrightarrow$ Before switch closed:

"short circuit"

Inductor:

$\hookrightarrow v_L(t) = L \frac{di_L(t)}{dt}$

$\hookrightarrow p_L(t) = v_L(t) \cdot i_L(t)$

$\hookrightarrow w_L(t) = \frac{1}{2} L i_L^2(t)$

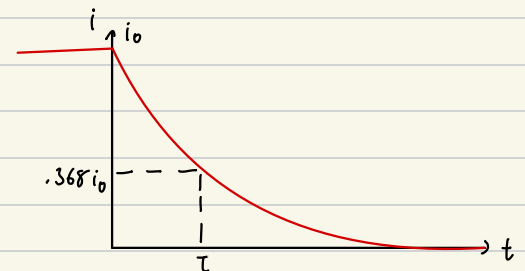
Resistor:

$\hookrightarrow v_R(t) = i_R(t) R = I_0 R e^{-t/\tau}$

$\hookrightarrow p_R(t) = v_R(t) i_R(t) = I_0^2 R e^{-2t/\tau}$

$\hookrightarrow w_R(t) = \int_0^t p_R(t) dt = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$

$\hookrightarrow \frac{1}{2} L I_0^2$ for large t .



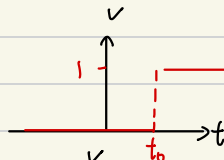
Strategy to solving source free RC, RL circuits

1. Find $i(0)$ for RL circuits, $v(0)$ for RC circuits
2. Determine V_0 or I_0 for particular element
3. Combine to form equivalent inductor / capacitor (if applicable)
4. Find R_{TH} (after switch is closed), hence τ

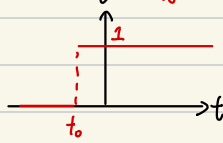
◦ Singularity functions

↳ Unit step, $u(t)$

$$u(t-t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

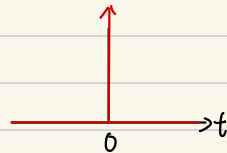


$$u(t+t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$



Side note:

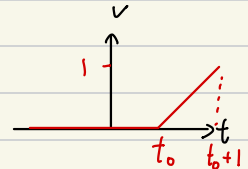
$$\frac{du(t)}{dt} = \delta(t) = \begin{cases} 0 & t < 0 \\ \infty & t = 0 \\ 0 & t > 0 \end{cases}$$



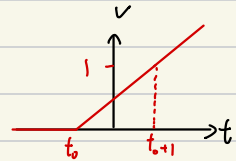
↳ This is known as the unit impulse function

↳ Unit ramp, $r(t)$

$$r(t-t_0) = \begin{cases} 0, & t \leq t_0 \\ t, & t \geq t_0 \end{cases}$$



$$r(t+t_0) = \begin{cases} 0, & t \leq -t_0 \\ t, & t \geq -t_0 \end{cases}$$



$$\hookrightarrow u(t) = \frac{dr(t)}{dt}$$

◦ Step response of RC/RL circuits:

↳ Natural response: $v_n = V_0 e^{-t/\tau}$ (RC) or $i_n = I_0 e^{-t/\tau}$ (RL)

↳ Energy stored in capacitor and inductor at $t=0$

Caused by "charged" up storage element releasing energy to circuit after switch is moved

↳ Forced response: $v_f = V_S (1 - e^{-t/\tau})$ (RC) or $i_f = I_S (1 - e^{-t/\tau})$ (RL)

↳ Caused by the independent sources in circuit after switch has moved.

↳ Transient response: $v_t = (V_0 - V_S) e^{-t/\tau}$ (RC) or $i_t = (i_0 - i_S) e^{-t/\tau}$ (RL)

↳ temporary part that dies out with time

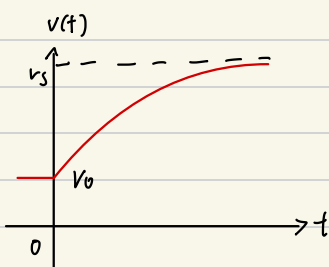
↳ Steady-state response: $v_{ss} = V_S$ (RC) or $i_{ss} = I_S$ (RL)

↳ Long time after external excitation is applied

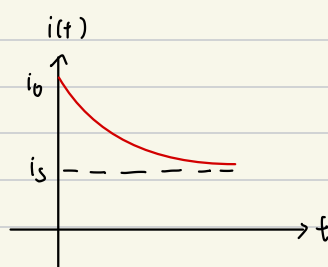
↳ Complete response: $v(t) = v_n + v_f = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau} = v_t + v_{ss}$ (RC)

$i(t) = i_n + i_f = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau} = i_t + i_{ss}$ (RL)

RC:



RL:



Strategy for solving First-order circuit

Step 1: Draw circuit for $t < 0s$, find initial inductor current $i_L(0+)$ or initial capacitor voltage $v_C(0+)$

Step 2: Draw circuit for $t = (0+)s$, replace inductor with current source $i = i_L(0+)$
replace capacitor with voltage source $v = v_C(0+)$

Then, find $v(0+)$ or $i(0+)$ as desired.

Step 3: Draw circuit $t > 0s$, determine R_{TH} of inductor/capacitor

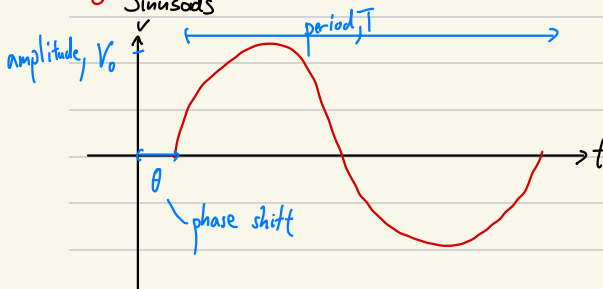
Find τ from this, $\tau = \frac{L}{R_{TH}}$ for RL , $\tau = R_{TH}C$ for RC

Step 4: Draw circuit $t \rightarrow \infty$, replace inductor with short circuit,
replace capacitor with open circuit

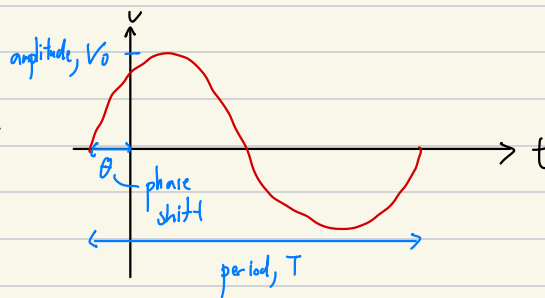
Find $i(\infty)$, $v(\infty)$

Thus, $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$ or $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

Sinusoids



$$v(t) = V_0 \sin(\omega t - \theta)$$



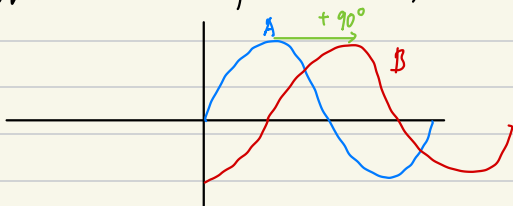
$$v(t) = V_0 \sin(\omega t + \theta)$$

$\hookrightarrow$ frequency, $f = \frac{1}{T}$ Hz
angular frequency, $\omega = 2\pi f = \frac{2\pi}{T}$

$\hookrightarrow$ Leading, lagging:

Leading: If A reaches peak before B, A leads B.

Lagging: If A reaches peak after B, A lags B.



$$V_A = V_0 \sin(\omega t)$$

V_A leads V_B by 90°

$$V_B = V_0 \sin(\omega t - 90^\circ)$$

V_B lags V_A by 90°

AC/DC

$\hookrightarrow$ DC: Constant current, voltage

$\hookrightarrow$ AC: Currents and voltage which vary with time. Typically periodic, sinusoidal.



Phasors

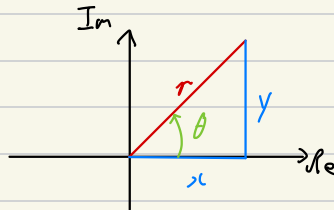
$\hookrightarrow z = x + jy, j = \sqrt{-1}$

$\hookrightarrow$ Forms:

$z = x + jy$, Rectangular form

$z = r \angle \theta$, Polar form

$z = r e^{j\theta}$, Exponential form (less used)



$x = r \cos \theta$

$y = r \sin \theta$

$r = \sqrt{x^2 + y^2}$

$\theta = \tan^{-1} \frac{y}{x}$

$\hookrightarrow$ Operations:

Add: $z_1 + z_2 = (x_1 + jy_1) + (x_2 + jy_2) = (x_1 + x_2) + j(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 + jy_1) - (x_2 + jy_2) = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 z_2 = r_1 r_2 \angle \theta_1 + \theta_2$

Division: $z_1 / z_2 = r_1 / r_2 \angle \theta_1 - \theta_2$

Reciprocal: $1/z = 1/r \angle -\theta$

Square root: $\sqrt{z} = \sqrt{r} \angle \theta/2$

Complex Conjugate: $z^* = x - jy = r \angle -\theta = r e^{-j\theta}$, $|z| = r = z z^*$

Negative phase: $z \angle -\theta = -z \angle -\theta + 180^\circ$

$\hookrightarrow$ Sinusoids as phasors:

$v(t) = V_0 \sin(\omega t + \theta) \rightarrow \bar{V} = V_0 \angle \theta - \frac{\pi}{2}$

$i(t) = I_0 \cos(\omega t + \theta) \rightarrow \bar{I} = I_0 \angle \theta$

★ Only when their frequency are equal.

$\hookrightarrow$ And $-j = \frac{1}{j}$, $j = 1 \angle 90^\circ$, $-j = 1 \angle -90^\circ$

$\hookrightarrow$ Multiplying a phasor by $j\omega \rightarrow$ Differentiating

Dividing a phasor by $j\omega \rightarrow$ Integrating

Impedance, Admittance

Phasor $\bar{Z}$ (Impedance) = $\bar{V} / \bar{I}$ (Voltage) / (Current):

$\hookrightarrow \bar{Z} = R(\omega) + jX(\omega)$ (Units Ω)

Phasor $\bar{Y}$ (Admittance) = $\bar{I} / \bar{V}$:

$\bar{Y} = G(\omega) + jB(\omega)$ (Units S)

$R(\omega)$: Resistance, $= \text{Re}\{\bar{Z}\}$

$X(\omega)$: Reactance, $= \text{Im}\{\bar{Z}\}$

$G(\omega)$: Conductance $= \text{Re}\{\bar{Y}\}$

$B(\omega)$: Susceptance $= \text{Im}\{\bar{Y}\}$

In General,

$G(\omega) \neq \frac{1}{R(\omega)}$

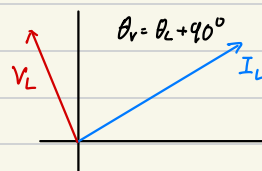
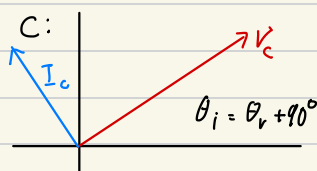
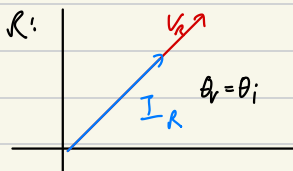
$B(\omega) \neq \frac{1}{X(\omega)}$

$\hookrightarrow$ Impedance of Resistor, Capacitor, Inductor:

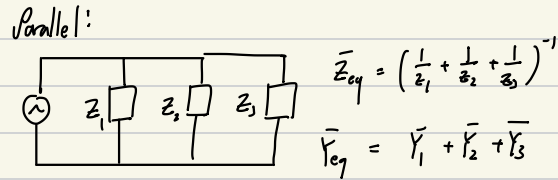
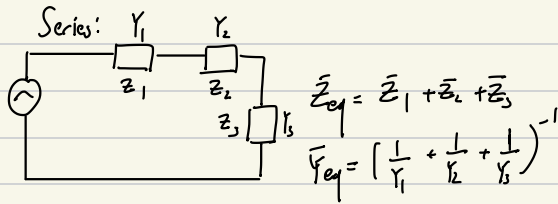
$\hookrightarrow \bar{Z}_R = V/I \angle 0^\circ$, purely real, voltage in phase with current

$\hookrightarrow \bar{Z}_C = V/I \angle -90^\circ = -j/\omega C = \frac{1}{\omega C} \angle -90^\circ$, purely imaginary, voltage lags current by 90°

$\hookrightarrow \bar{Z}_L = V/I \angle 90^\circ = j\omega L = \omega L \angle 90^\circ$, purely imaginary, voltage leads current by 90°



Series / Parallel Impedance, Admittance

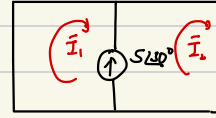


KCL and KVL in phasor domain

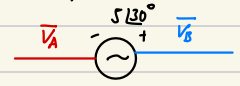
→ KVL: $\sum_{i=1}^n V_{rise} = \sum_{i=1}^n V_{drop}$

KCL: $\sum_{i=1}^n \bar{I}_{into\ node} = \sum_{i=1}^n \bar{I}_{out\ of\ node}$

Supermesh:



Supernode:



→ This means we can use nodal / mesh analysis in the phasor domain → Treat supernode, supermesh as previously.

Superposition in phasor domain

→ Still holds, pay attention to ω , might be different.

$$V(t) = V_{01}(\cos \omega_1 t + \theta_1) + V_{02}(\cos \omega_2 t + \theta_2) \text{ etc.}$$

Thevenin, Norton equivalents

→ Still the same,

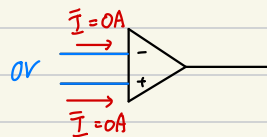
$$\bar{V}_{TH} = \bar{I}_N \bar{Z}_{TH} \text{ (or } \bar{Z}_N)$$

Op-Amp and steady state AC

→ What would've thought, still the same.

1. No current enters input terminals

2. Voltage across input terminal is zero



Strategy to AC steady-state circuit analysis

1.) Convert sources to phasors, either all sin, in such case, $\cos \omega t = \sin(\omega t + 90^\circ)$ or all cos, in such case, $\sin \omega t = \cos(\omega t - 90^\circ)$

2.) Convert passive elements into phasor, using ω

$$\bar{Z}_R = R$$

$$\bar{Z}_L = j\omega L$$

$$\bar{Z}_C = -j/\omega C$$

3.) Solve it yourself.

4.) Convert back into original sinusoids, using ω

Instantaneous power

$$p(t) = v(t) i(t)$$

$$= V_0 I_0 \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$= \underbrace{\frac{1}{2} V_0 I_0 \cos(\theta_v - \theta_i)}_{\text{avg power}} + \underbrace{\frac{1}{2} V_0 I_0 \cos(2\omega t + \theta_v + \theta_i)}_{\text{oscillating power}}$$

Average Power (aka Real Power)

$$P_{\text{avg}} \text{ (or } P) = \frac{1}{2} |\bar{V}| |\bar{I}| \cos(\theta_v - \theta_i) \quad (\text{Purely Real})$$

$$= \frac{1}{2} |\bar{V}| |\bar{I}| \cos \theta_z$$

$$= \frac{1}{2} \frac{|\bar{V}|^2}{|\bar{Z}|} \cos \theta_z$$

$$= \frac{1}{2} |\bar{Z}| |\bar{I}|^2 \cos \theta_z$$

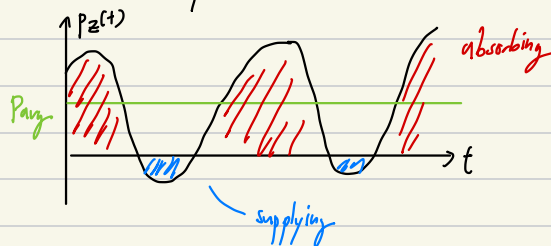
$$= \frac{1}{2} \operatorname{Re} \{ \bar{V} \bar{I}^* \}$$

$$= S \cos(\theta_z) \text{, see below}$$

↳ Resistive load (R) absorbs power all times

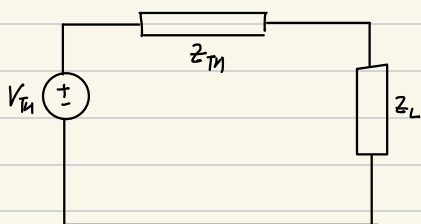
↳ Reactive load (L or C) absorbs zero average power

↳ General impedance:



Maximum average power

$$\text{Maximum average power, } P_{\text{max}} = \frac{|\bar{V}_{\text{Th}}|^2}{8R_{\text{Th}}}, \text{ where } \bar{Z}_{\text{load}} = \bar{Z}_{\text{Th}}^* = R_{\text{Th}} - jX_{\text{Th}}$$



Root-mean-square value

↳ For any periodic function:

↳ For sinusoids:

$$X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

$$X_{\text{rms}} = \frac{X_0}{\sqrt{2}}$$

$$\longrightarrow \begin{aligned} I_{\text{rms}} &= I_{\text{eff}} = \frac{I_0}{\sqrt{2}} \\ V_{\text{rms}} &= V_{\text{eff}} = \frac{V_0}{\sqrt{2}} \end{aligned}$$

$$\text{where } P_{\text{avg}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} \text{ for a purely resistive load.}$$

Apparent Power, power factor, reactive power

$$\text{Apparent Power, } S = \frac{1}{2} V_{\text{peak}} I_{\text{peak}} = V_{\text{rms}} I_{\text{rms}} = |\bar{S}| \quad (\text{Units: VA, Volt-Ampere})$$

$$\text{Reactive power, } Q \quad (\text{Units: VAR})$$

$$= \operatorname{Im} \{ \bar{S} \}$$

Technically not Watts

$$= S \sin(\theta_z)$$

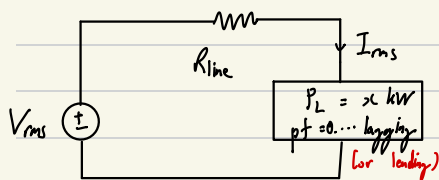
$$\text{Average Power, } P = S \cos(\theta_z) = \operatorname{Re} \{ \bar{S} \}$$

$$\text{Power Factor, pf} = P/S = \cos \theta_z$$

↳ leading if current leads voltage, capacitive load, $\theta_v - \theta_i = \theta_z < 0$

↳ lagging if current lags voltage, inductive load, $\theta_v - \theta_i = \theta_z > 0$

Power losses



Steps:

1) Find S_L

$$S_L = \frac{P_L}{\text{pf}} = V_{rms} I_{rms}$$

2) Use S_L to determine I_{rms} or V_{rms}

$$I_{rms} = \frac{S_L}{V_{rms}} \quad , \quad V_{rms} = \frac{S_L}{I_{rms}}$$

3) Determine P_{losses} via:

$$\begin{aligned} P_{losses} &= |I_{rms}|^2 R_{line} \\ &= \frac{P_L^2}{V_{rms}^2 (\text{pf})^2} R_{line} \end{aligned}$$



Complex Power

Complex Power, $\bar{S} = (\text{Average power}) + j(\text{Reactive Power})$ (Units: VA)

$$= P + jQ \quad (P: W, Q: VAR)$$

Blue $\rightarrow$ Average (or Real) Power

$$= V_{rms} I_{rms} \cos(\theta_2) + j V_{rms} I_{rms} \sin(\theta_2)$$

Green $\rightarrow$ Reactive power

$$= S \cos(\theta_2) + j S \sin(\theta_2)$$

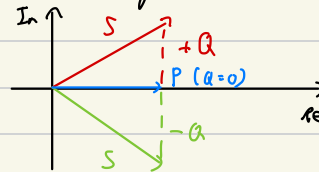
$$= \bar{V}_{rms} (\bar{I}_{rms})^*$$

$$= |\bar{V}_{rms}|^2 / \bar{Z}_L^*$$

$$= |\bar{I}_{rms}|^2 \bar{Z}_L$$

Magnitude of complex power $|\bar{S}| = S$ = apparent power

Power triangle:



$Q=0 \rightarrow$ resistive loads, unity pf

$Q < 0 \rightarrow$ capacitive loads, leading pf

$Q > 0 \rightarrow$ inductive loads, lagging pf

Conservation of complex power

$$\bar{S}_{tot} = \bar{S}_1 + \bar{S}_2 + \bar{S}_3 + \dots + \bar{S}_N \quad \text{for } N \text{ elements}$$

Power factor correction

To increase pf from $\text{pf}_1 = \cos \theta_1$ to $\text{pf}_2 = \cos \theta_2$, given real power P , a load needs a capacitance:

$$C = \frac{P(\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

o Summary of Power (50 formulas in a row)

Quick definitions: $\theta_z = \theta_v - \theta_i = \theta_{rms} - \theta_{rms}$, $V_{rms} = V_0 / \sqrt{2}$, $I_{rms} = I_0 / \sqrt{2}$, $\bar{V}_{rms} = \bar{V} / \sqrt{2}$, $\bar{I}_{rms} = \bar{I} / \sqrt{2}$

↳ Instantaneous Power, *function of time*

$$p(t) = v(t)i(t) = \frac{1}{2} V_0 I_0 \cos(\theta_v - \theta_i) + \frac{1}{2} V_0 I_0 \cos(2\omega t + \theta_v + \theta_i)$$

avg *oscillating*

↳ Complex power, *phasor*

$$\begin{aligned} \bar{S} &= \frac{1}{2} \bar{V} \bar{I}^* \\ &= \bar{V}_{rms} \bar{I}_{rms}^* \\ &= |\bar{V}_{rms}|^2 / \bar{Z}_L^* \\ &= |\bar{I}_{rms}|^2 \bar{Z}_L \\ &= V_{rms} I_{rms} \angle \theta_z = |\bar{V}_{rms}| |\bar{I}_{rms}| \angle \theta_z \\ &= V_{rms} I_{rms} \cos(\theta_z) + j V_{rms} I_{rms} \sin(\theta_z) \\ &= S \cos(\theta_z) + j S \sin(\theta_z) \\ &= P + j Q \quad (\text{Volts-Ampere}) \end{aligned}$$

↳ Apparent power, *scalar*

$$\begin{aligned} S &= |\bar{S}| \\ &= \frac{1}{2} V_0 I_0 \quad (V_0, I_0 \text{ are amplitudes or peaks}) \\ &= V_{rms} I_{rms} = |\bar{V}_{rms}| |\bar{I}_{rms}| \\ &= \sqrt{P^2 + Q^2} \quad (\text{Volts-Ampere}) \end{aligned}$$

↳ Power factor, *scalar*

$$\begin{aligned} \text{pf} &= \cos \theta_z \\ &= P/S \quad (\text{unitless}) \end{aligned}$$

↳ Real (or average) power, *scalar*

$$\begin{aligned} P &= \frac{1}{2} V_0 I_0 \cos(\theta_z) \\ &= V_{rms} I_{rms} \cos(\theta_z) = |\bar{V}_{rms}| |\bar{I}_{rms}| \cos \theta_z \\ &= S \cos(\theta_z) \quad \cos \theta_z = \text{pf} \\ &= \text{Re}\{\bar{S}\} \quad (\text{Watts}) \end{aligned}$$

↳ Reactive power, *scalar*

$$\begin{aligned} Q &= \frac{1}{2} V_0 I_0 \sin(\theta_z) \\ &= V_{rms} I_{rms} \sin(\theta_z) = |\bar{V}_{rms}| |\bar{I}_{rms}| \sin \theta_z \\ &= S \sin(\theta_z) \\ &= \text{Im}\{\bar{S}\} \quad (\text{Volts-Ampere-Reactive}) \end{aligned}$$

↳ Maximum average power
When $\bar{Z}_L = \bar{Z}_{TH}^* = R_{TH} - jX_{TH}$

$$P_{\max} = |\bar{V}_{TH}|^2 / 8 R_{TH}$$

↳ Loads and power:

↳ pf correction

Purely resistive loads:

$$\left. \begin{aligned} Q &= 0 \\ P &= V_{rms} I_{rms} \end{aligned} \right\} \bar{S} = P$$

$$\text{pf} = 1 \quad \text{"unity pf"}$$

Purely capacitive loads:

$$\left. \begin{aligned} P &= 0 \\ Q &= -V_{rms} I_{rms} \end{aligned} \right\} \bar{S} = -j V_{rms} I_{rms}$$

$$\text{pf} = 0 \quad \text{"leading"} \\ \text{voltage lags current by } 90^\circ$$

Purely inductive loads:

$$\left. \begin{aligned} P &= 0 \\ Q &= V_{rms} I_{rms} \end{aligned} \right\} \bar{S} = j V_{rms} I_{rms}$$

$$\text{pf} = 0 \quad \text{"lagging"} \\ \text{voltage leads current by } 90^\circ$$

To change $\text{pf}_1 = \cos \theta_1 \rightarrow \text{pf}_2 = \cos \theta_2$
with given Avg Power P ,

$$C = \frac{P(\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

Inductive loads:

$$Q > 0$$

$$\text{pf} \rightarrow \text{lagging} > 0$$

current lags voltage, $\theta_z > 0$

Capacitive loads:

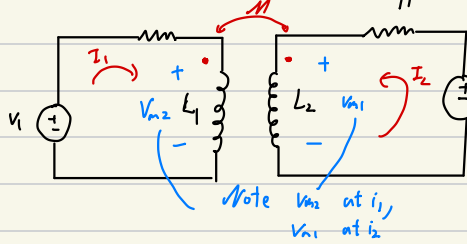
$$Q < 0$$

$$\text{pf} \rightarrow \text{leading} > 0$$

current leads voltage, $\theta_z < 0$

o Mutual inductance

↳ Mutual flux b/w coil with applied coil, applied current, second coil



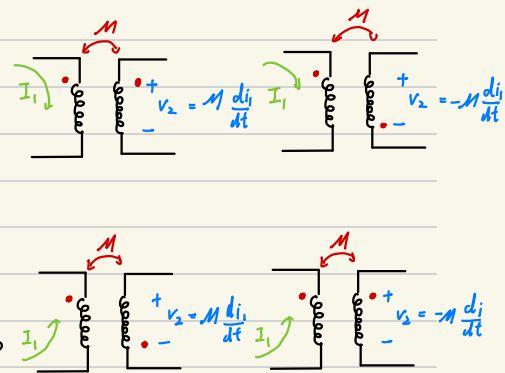
Mutual inductance, $M = M_{21} = M_{12} = k\sqrt{L_1 L_2} \rightarrow k = \text{Coupling coeff. (ideal for } k=1)$

Phasor = $M \rightarrow \bar{V}_m = j\omega M$
 $V_2 = M \frac{dI_1}{dt} \rightarrow \bar{V}_2 = j\omega M \bar{I}_1$
 $V_1 = M \frac{dI_2}{dt} \rightarrow \bar{V}_1 = j\omega M \bar{I}_2$

↳ Dot convention

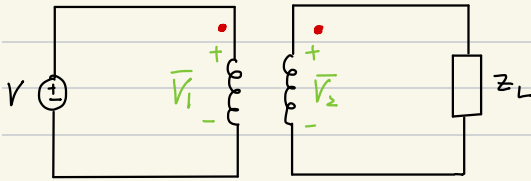
Current 1, I_1 enters through dot $\rightarrow$ Voltage 2, V_2 +ve at dot

Current 1, I_1 leaves from dot $\rightarrow$ Voltage 2, V_2 -ve at dot



o Ideal transformers

Diagram: $\bar{I}_1 \rightarrow 1:n \rightarrow \bar{I}_2$

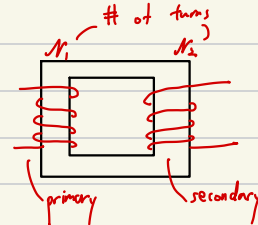


Equations:

Turns ratio, $n = \frac{N_2}{N_1}$

$\frac{V_2}{V_1} = \pm n = \pm \frac{N_2}{N_1}$

$\frac{I_2}{I_1} = \mp \frac{1}{n} = \mp \frac{N_1}{N_2}$



Dot convention:

If $\bar{V}_1, \bar{V}_2$ both +ve or both -ve at dot $\rightarrow \bar{V}_2 / \bar{V}_1 = +n = +N_2 / N_1$

else $\rightarrow \bar{V}_2 / \bar{V}_1 = -n = -N_2 / N_1$

If $\bar{I}_1, \bar{I}_2$ both enter or leave the dot $\rightarrow \bar{I}_2 / \bar{I}_1 = -1/n = -N_1 / N_2$

else $\rightarrow \bar{I}_2 / \bar{I}_1 = +1/n = +N_1 / N_2$

↳ Conservation of complex power in ideal transformers

$\bar{S}_1 = \bar{V}_1 \bar{I}_1^* = \bar{V}_2 \bar{I}_2^* = \bar{S}_2$

↳ Impedance "matching" $\rightarrow Z_{in} = Z_L / n^2 = \bar{V}_1 / \bar{I}_1 = \frac{1}{n^2} \frac{\bar{V}_2}{\bar{I}_2}$

o Types of transformers

Optimus Prime

- Step-up: $n > 1$, secondary voltage > primary voltage
- Step-down: $n < 1$, secondary voltage < primary voltage
- Isolation: $n = 1$, secondary voltage = primary voltage

Assumption of ideal transformers:

1. $L_1, L_2, M \rightarrow \infty$ (Large reactances)
2. $k = 1$ (Coupling coeff = 1)
3. $R_1 = 0 = R_2$ (0 loss of flux from core)

